

Preprocessing and Visualizing Data

Ordered observations, graphical integrity, and amounts

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Continuing into Week 3

- Bring the cleaned data, data dictionary, and decision record from Lab 02.
- Practical work continues in the scheduled Class B meeting.
- Use R or Python. The notes include complete setup for each example.
- Check the course repository for the current instructions on Lab 03 and Assignment 01.

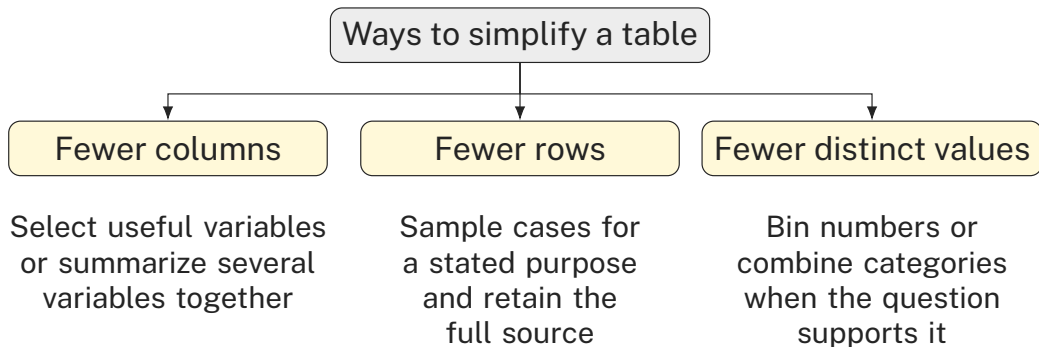
Keep assessed work private until the deadline and instructor release.

From Preparation to Visual Questions

Week 2 checked values, types, units, and transformations.
Today we add meaningful context and choose plots that help us examine the prepared data.



Ways to Simplify a Table



Reducing detail is a decision to justify. Keep the original data and describe what the reduced view omits.

Simple Random and Stratified Samples

Population, 32 cases
12 A, 20 B



Case IDs run left to right, then down.
Gold rings identify selected cases.

Stratified sample, 3 A + 5 B



1 5 9 13
17 21 25 29

Simple random sample, 2 A + 6 B



2 8 14 18
22 26 30 32

Both designs select eight cases without replacement. The displayed selections illustrate the designs; software seeds need not select these same IDs.

PCA as a Change of View

A principal component combines existing numeric variables. The first points along the direction of greatest variation in the chosen scale.

- Keeping fewer components can make a complex table easier to view.
- Some detail is lost, and the new axes may be harder to interpret than the originals.

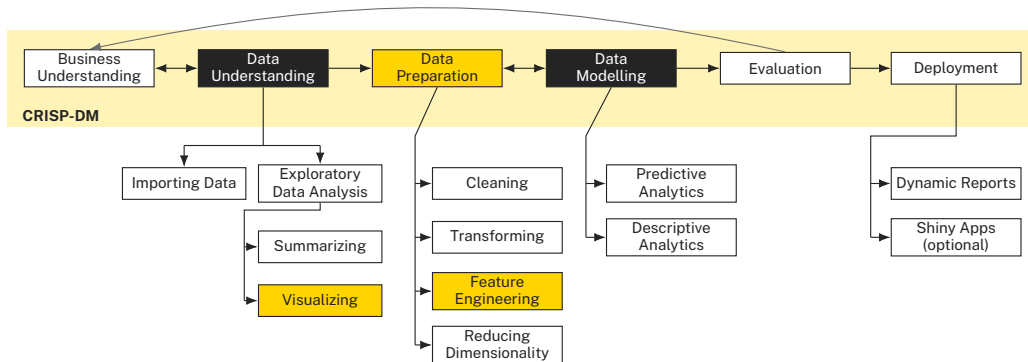
This continues the optional Week 2 example. It is not a prerequisite for Lab 03.

Two Ways to Explore PCA

- [Powell and Lehe, PCA explained visually](#). Explore how the direction of greatest variation changes as the points move.
- [Jaadi, A step-by-step PCA explanation](#). Optional mathematical background after the geometric idea is clear.

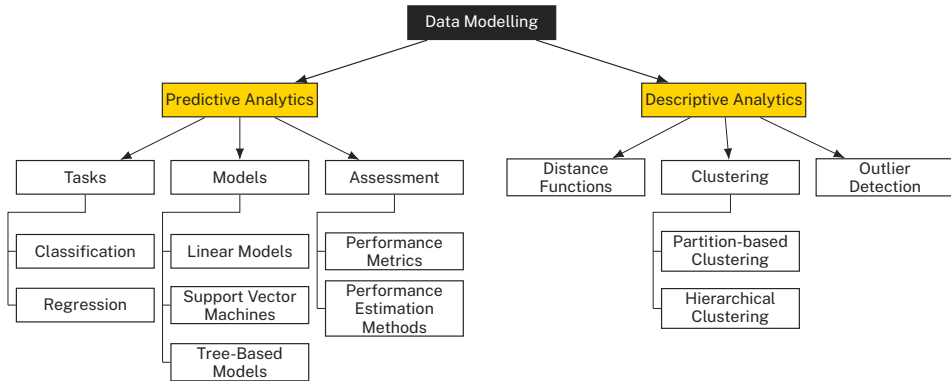
Begin with the geometry. Matrix calculations are further reading, not required.

Features and Visual Exploration



We move between preparation and exploration as the data raises new questions.

Where These Choices Lead



Orientation only. Model fitting, evaluation, and clustering are taught in later weeks.

What You Will Practise

- Explain how lag columns record the order of observations.
- Calculate relative changes when the baseline has a meaningful nonzero value.
- Use plots and numeric summaries to check each other.
- Choose, label, and improve charts of amounts.

Class B develops distributions, proportions, and associations in more detail.

Give a New Variable a Meaning

A useful derived column expresses a relationship that matters to the question.

Examples include an amount per person, a previous measurement, or a change from one interval to the next.

Record the Context of a Measurement

- A sensor reading may depend on what was measured a minute earlier.
- Repeated measurements from one person share a subject identity.
- Nearby places may share conditions. A location code alone does not define a neighbourhood.

State what each row represents before deriving a contextual variable.

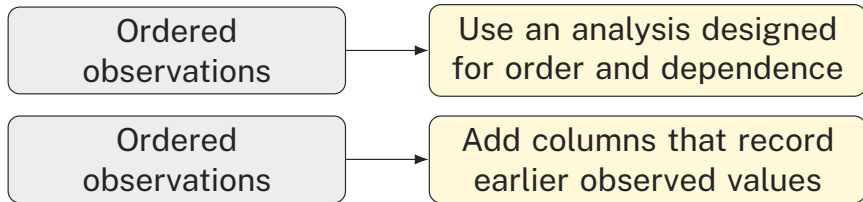
Order and Identity Matter

Columns record measured variables
Rows identify observations

Time order and subject
identity are part of the data's
meaning. Keep both explicit.

A row is an observation, not a promise of independence. Shuffling a time series
can hide the very pattern we want to examine.

Two Complementary Approaches



The new columns
preserve context.
Dependence can remain.

Today we construct context columns. Choosing an appropriate model comes later.

A Four-Column Time Delay Embedding

One subject has seven equally spaced, chronologically ordered readings, 4, 5, 6, 7, 8, 9, 10.

Time t	$Y(t)$	$Y(t - 1)$	$Y(t - 2)$	$Y(t - 3)$
4	7	6	5	4
5	8	7	6	5
6	9	8	7	6
7	10	9	8	7

Each row contains the current value and its three preceding values. The first three times lack enough history, leaving four complete rows.

Build the Lag Columns

```
# One subject, equally spaced times, already in chronological order.  
y <- c(4, 5, 6, 7, 8, 9, 10)  
lags <- embed(y, dimension=4)  
colnames(lags) <- c("Y", "lag1", "lag2", "lag3")  
print(lags)  
# Rows 7,6,5,4; 8,7,6,5; 9,8,7,6; 10,9,8,7.
```

embed retains the numeric type of this input. Here the matrix stores double-precision values.

Build the Same Lag Columns

Start with the imports in the notes.

```
# One subject, equally spaced times, already in chronological order.
y = pd.Series([4, 5, 6, 7, 8, 9, 10], dtype=float)
lags = pd.DataFrame({
    "Y": y, "lag1": y.shift(1),
    "lag2": y.shift(2), "lag3": y.shift(3),
}).dropna()
print(lags.to_numpy())
# Rows [7,6,5,4], [8,7,6,5], [9,8,7,6], [10,9,8,7].
```

Sort within each subject first. A row shift represents a fixed duration only when measurements are equally spaced.

Distinguish an Amount from a Rate of Change

For an amount with a meaningful ratio scale and a nonzero previous value, the fractional relative change is

$$r_t = \frac{Y_t - Y_{t-1}}{Y_{t-1}}.$$

Multiply by 100 to express the fractional change as a percentage.

A zero or missing previous value requires an explicit handling rule.

$$6 \longrightarrow 9$$

$$(9 - 6)/6 = 0.5 = 50\%$$

Constant Proportional Growth

```
y <- 4 * 1.5^(0:5)
stopifnot(all(!is.na(y)), all(y > 0))
change <- diff(y) / head(y, -1)
print(change) # Five values equal to 0.5.
print(100 * change) # 50 percent each interval.
```

This exact sequence grows by 50% each interval. Relative change is constant here; that does not make arbitrary time series independent or trend-free.

Align Changes with Their Times

Continue from the imports in the notes.

```
y = pd.Series(4 * 1.5 ** np.arange(6))
assert y.notna().all() and (y > 0).all()
change = y.pct_change(fill_method=None)
print(change.tolist())
# Missing first value, then five values equal to 0.5.
print((100 * change).tolist())
# Missing first value, then 50 percent each interval.
```

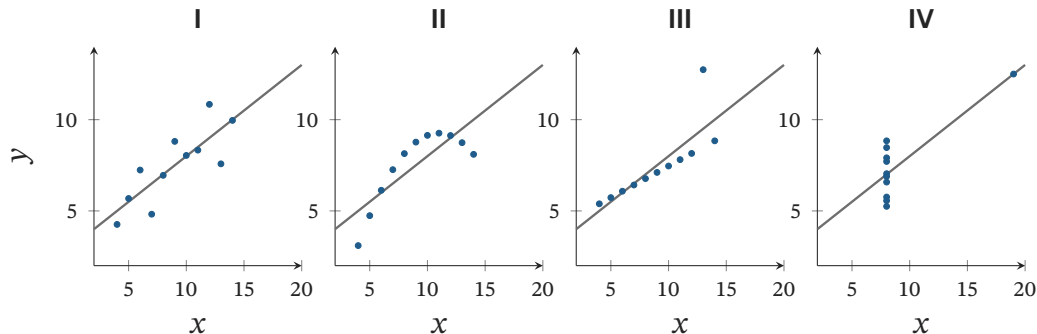
R returns five changes. pandas retains six time positions, with the first change missing.

Look at the Values as Well as the Summaries

A mean, a standard deviation, and a correlation describe selected features of a dataset. A plot can reveal what those numbers leave out.



Anscombe's Quartet



Each panel contains the original eleven points. The grey line $y = 3 + 0.5x$ is the rounded common fit, shown for comparison.[1]

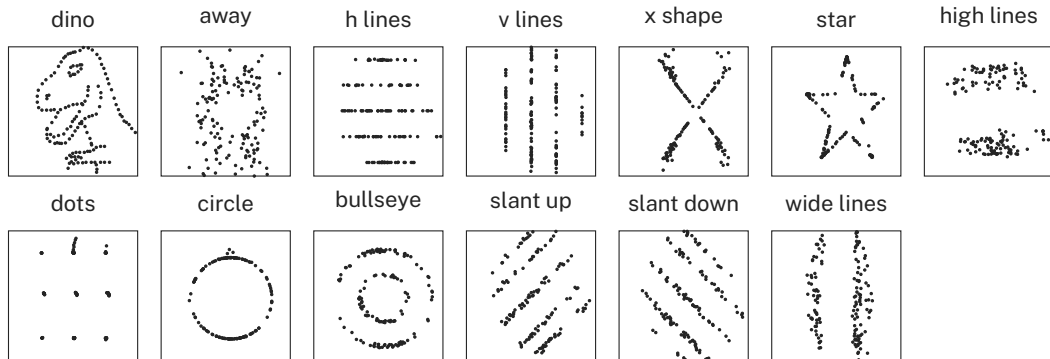
Similar Summaries, Different Patterns

Summary	Rounded value in each set
Mean x ; sample SD x	9.00; 3.32
Mean y ; sample SD y	7.50; 2.03
Pearson correlation r	0.82
Approximate fitted line	$y = 3 + 0.5x$

- I shows scatter around a straight pattern; II shows curvature.
- III has an unusual point above an otherwise near-linear pattern.
- IV has little variation in x except for one point.

The summaries agree at the displayed precision. They are not identical at every precision.

The Datasaurus Dozen



Thirteen authentic datasets, 142 points each, on common 0–100 axes. The original dinosaur is by Alberto Cairo; the other shapes are by Matejka and Fitzmaurice.[2]

Nearly Equal Numbers, Distinct Shapes

Summary	Range across 13 sets
Mean x	54.26015 to 54.26993
Mean y	47.83025 to 47.83983
Sample SD x	16.76001 to 16.77000
Sample SD y	26.93000 to 26.94000
Pearson r	-0.06945 to -0.06034

These ranges are calculated from the original 1846 coordinates. The small negative correlations describe linear association; they do not rule out a clear nonlinear pattern.

Use Both Calculations and Graphs

- Ask what a summary measures and what it ignores.
- Inspect shape, gaps, subgroups, and unusual observations.
- Return to the data dictionary before treating an unusual point as an error.
- Report a pattern as evidence about these data, with its limits.

The notes reproduce both published examples in R and Python and link directly to the Datasaurus data.

Amounts and Distributions

Amounts, this class

Bar and dot plots

Grouped and stacked bars

Heatmaps

Distributions, Class B

Histograms and density plots

Boxplots and violin plots

ECDF plots

Choose from the question, variable type, units, and meaningful order. The same numeric column may support different questions.

Proportions and Associations

Proportions

Part of a whole
Pie and stacked
bars

Two variables

Association or
change
Scatterplots
Line plots for order

Several variables

Patterns across
columns
Scatterplot
matrices
Parallel coordinates

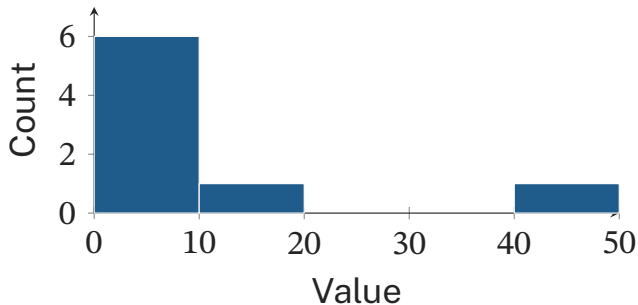
These families are developed in Class B. A line joins observations only when that connection has meaning.

Four Libraries Make Different Initial Choices

Interface	Default bin choice
R hist	Sturges rule followed by convenient break locations.
ggplot2 geom_histogram	30 bins if no bin choice is supplied.
Matplotlib hist	10 bins under the standard configuration.
seaborn histplot	NumPy's auto bin rule.

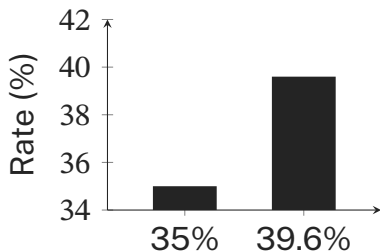
Defaults are starting points. For a deliberate cross-language comparison, use the same observed values, explicit edges, and missing-value policy.

A Reproducible Histogram Comparison



Values 2, 4, 4, 5, 7, 9, 12, 41 and one missing entry. Explicit edges 0, 10, 20, 30, 40, 50 give counts 6, 1, 0, 0, 1 in both languages. The complete code is in the notes.

What Does a Clipped Bar Imply?

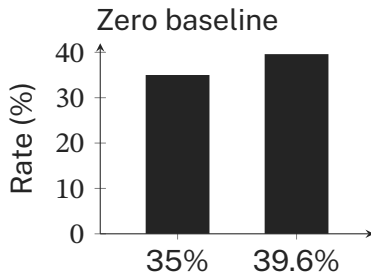
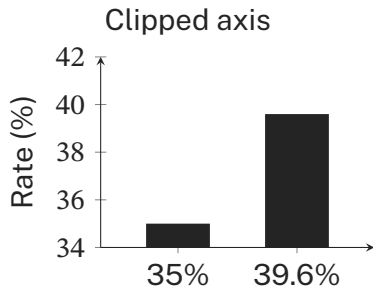


A teaching redraw compares rates of 35% and 39.6%.

The difference is 4.6 percentage points. Relative to 35%, the increase is about 13.1%.

The visible bar lengths start at 34%, making the second appear 5.6 times as long.

Show the Same Values from Zero



Both show an increase. A zero baseline makes bar length proportional to the rate being compared.

Magnitude and Labels

Match visual magnitude to numeric magnitude.

Label the comparison completely.

For bars, begin at zero so length represents the stated amount.

Give the variable, unit, population, time period, and source.

Adapted as course design principles from Tufte.[3]

Consistency and Context

Show variation in data, not decoration.

Give enough context to interpret the result.

Keep scales and encodings consistent across comparable panels.

Report exclusions and avoid selecting a range that conceals the pattern.

A clear chart makes the comparison easier to inspect, including the choices that shaped it.

Choose an Encoding for the Task

Compare numeric values

Aligned position and length usually support more precise comparisons than area, angle, or intensity.

Identify groups

Use hue, shape, line style, or text to distinguish groups. Categories do not need a magnitude ranking.

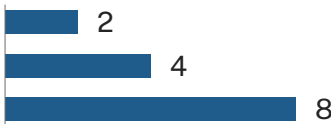
A perceptual comparison of quantities is a different task from identifying which group a mark belongs to.

Compare the Same Three Values

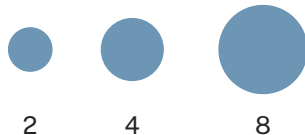
Aligned positions



Lengths from the same zero



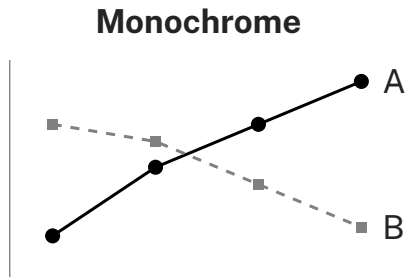
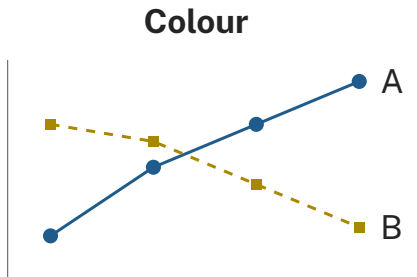
The same values encoded by area



The area of the last circle is four times that of the first. Comparing that ratio is harder than comparing aligned lengths.

Area and angle can communicate magnitude, but aligned position or length usually supports more precise comparisons.

Keep Meaning When Colour Changes



A and B remain identifiable through direct labels, circles versus squares, and solid versus dashed lines. This is a monochrome check, not a simulation of a particular visual condition.

What Should the Reader Compare?

Use bars or dots to compare amounts across categories. Use grouped or stacked bars when the question includes both categories and their components.



Start from Data, Marks, and Labels

- Data, five teams' regular-season wins in the Atlantic Division.
- Mapping, team on the category axis and wins on the numeric axis.
- Mark, a bar or a dot for each team.
- Labels and scales, season, unit, readable names, and a deliberate range.

The full twenty-value table and both language setups are in the notes before the plotting code.

Express the Same Design in Either Language

R

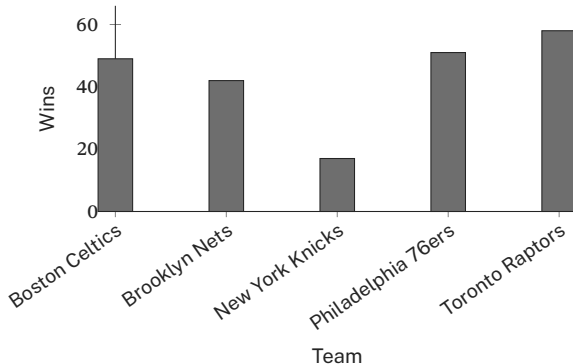
`ggplot` specifies data and mappings. A layer such as `geom_col` supplies the marks. Labels, scales, and a theme complete the plot.

These are two implementations of the same visual choices. Python also has layered interfaces; no language owns the underlying idea.

Python

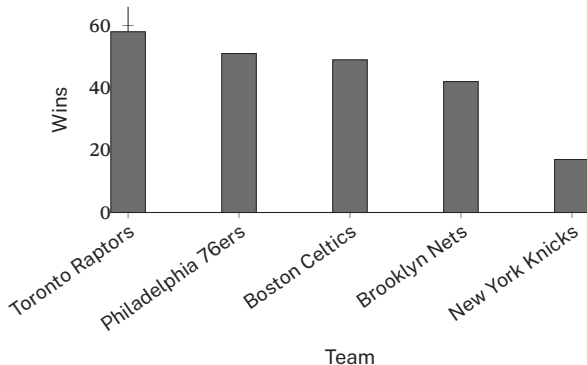
A Matplotlib figure contains axes. `ax.barh` supplies bars on those axes. Axis methods add the labels, limits, and title.

Alphabetical Order with Rotated Labels



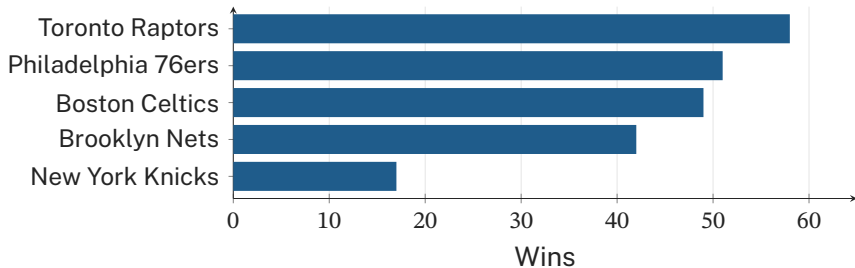
Alphabetical order supports lookup. The long team names fit only when rotated, so each one is read at an angle.

Order by the Quantity You Compare



Ordering by wins supports ranking, but the names are still rotated. Keep inherent order for ordinal categories or time.

Turn the Bars to Make Names Readable



Toronto has 58 wins and New York has 17 in 2018–19. The largest value appears at the top and all names fit.

Build an Ordered Horizontal Bar Chart

Continue from the NBA setup in the notes.

```
p <- ggplot(ordered, aes(x=`2018-19`, y=Team)) +  
  geom_col(fill=blue) + xlim(0, 65) +  
  labs(x="Wins", y=NULL, title="Atlantic Division, 2018-19") +  
  theme_minimal()  
ggsave("class-05-bars-r.pdf", p, width=7, height=4)
```

ordered stores the teams as factor levels from smallest to largest; the largest level appears at the top.

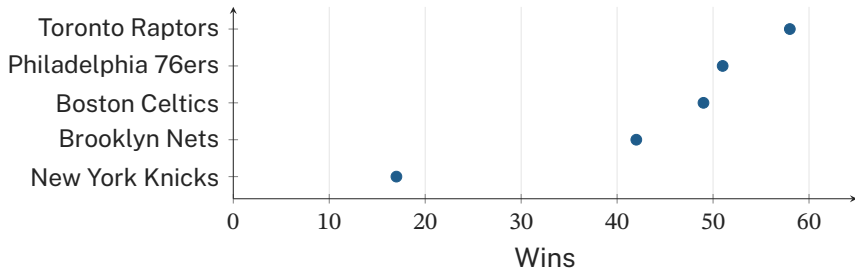
Keep the Same Category Order

Continue from the NBA setup in the notes.

```
fig, ax = plt.subplots()
ax.barh(ordered["Team"], ordered["2018-19"], color=blue)
ax.set(xlabel="Wins", title="Atlantic Division, 2018-19", xlim=(0, 65))
fig.tight_layout()
fig.savefig("class-05-bars-python.pdf")
plt.close(fig)
```

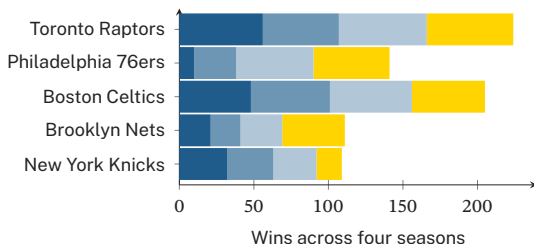
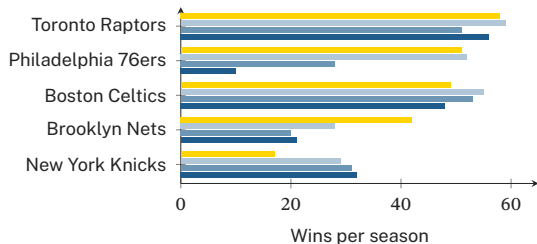
Matplotlib places the first category at the bottom here. The ascending data order gives the same top-to-bottom ranking as R.

Compare Positions on a Common Scale



Dots represent the same five wins. Position carries the comparison, so a carefully labelled narrower range can be useful when values are close. This view keeps the same 0–65 scale as the bars.

Grouped and Stacked Bars Differ



2015-16

2016-17

2017-18

2018-19

Grouped bars compare seasons from zero. Stacked bars compare four-season totals and their components.

Control Grouping and Stacking

Continue from the long-table setup in the notes.

```
p <- ggplot(nba_long, aes(x=Wins, y=Team, fill=Season)) +  
  geom_col(position=position_stack(reverse=TRUE), width=.7) +  
  scale_fill_manual(values=colors) + xlim(0, 240) +  
  labs(x="Wins across four seasons", y=NULL) + theme_minimal()  
ggsave("class-05-stacked-r.pdf", p, width=8, height=5)
```

`reverse=TRUE` puts seasons left to right in chronological order. The grouped example instead uses `position_dodge`.

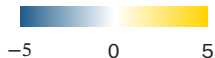
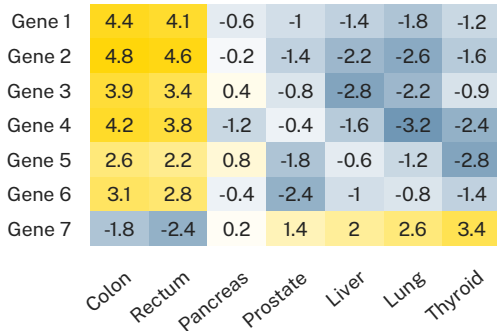
Stack the Same Four Seasons

Continue from the wide-table setup in the notes.

```
ax = wide.plot.barh(stacked=True, color=colors, width=0.7)
ax.set(xlabel="Wins across four seasons", xlim=(0, 240))
ax.legend(title="Season", ncols=2, loc="upper left", bbox_to_anchor=(1, 1))
ax.figure.tight_layout()
ax.figure.savefig("class-05-stacked-python.pdf")
plt.close(ax.figure)
```

Each season is one column in wide. Omitting `stacked=True` produces grouped bars.

Read a Numeric Matrix Through Colour



Deviation from
a chosen
reference.

Arbitrary units.

Synthetic
values.

Synthetic classroom values, printed inside each cell. The tissue names illustrate an RNA-analysis layout; these are not measured expression data or row-standardized scores.

A Colour Scale Does Not Require Clustering

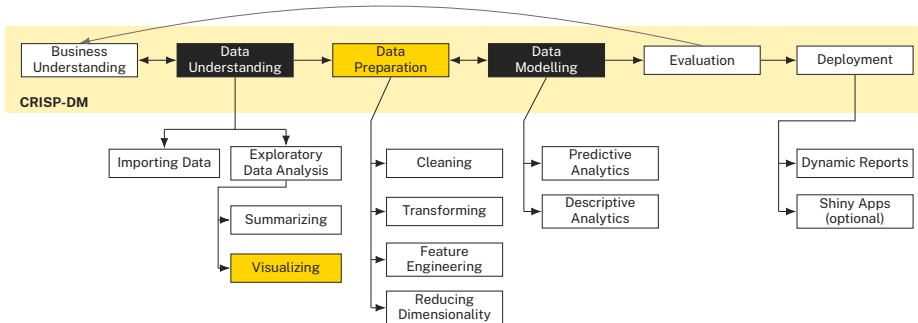
- Each cell maps one numeric value to a colour. White is fixed at the reference value zero in this example.
- Rows and columns retain the stated order. The notes include the complete matrix and both plotting implementations.
- A clustered heatmap can reorder similar rows or columns and add a tree. The result depends on the similarity and grouping rules.

Clustering methods come later in the course. Reading the cells and their scale is the task here.

Check Meaning, Evidence, and Presentation

- Derived variables need a clear definition and valid inputs.
- Lag columns preserve history; relative changes need a meaningful baseline.
- Plots and summaries describe different aspects of the same data.
- Make the question, amount, unit, order, and grouping visible.
- Use readable labels and more than colour to identify groups.

Continue with Distributions and Associations



Class B develops the remaining chart families and applies these principles in Lab 03.

Continue with the Examples That You Need

- [Wilke, Visualizing amounts](#). Ordering, orientation, and alternatives to bars.
- [Matejka and Fitzmaurice, Same Stats, Different Graphs](#). Published data behind the thirteen shapes.
- [W3C, Use of Color](#). Pair colour with text, shape, or another cue.
- [ggplot2, Bar charts](#) or [Matplotlib, Horizontal bars](#). Read the examples for your chosen language.

Sources Cited in This Class

- [1] F. J. Anscombe. “Graphs in statistical analysis”. In: *The American Statistician* 27.1 (1973), pp. 17–21.
- [2] J. Matejka and G. Fitzmaurice. “Same stats, different graphs: generating datasets with varied appearance and identical statistics through simulated annealing”. In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 2017, pp. 1290–1294.
- [3] E. R. Tufte. *The Visual Display of Quantitative Information*. Graphics Press.